

AI335L Deep Learning Lab
Experiment No. 12 — Phase 5

Skin Cancer Detection

Using Deep Learning

Phase 5 Evaluation Report

Rigorous Evaluation: Ablations, Errors, and Limitations

Course	AI335L Deep Learning Lab
Experiment	Experiment No. 12 — Phase 5
Project	Skin Cancer Detection using Transfer Learning
Final Model	Transfer ResNet-18 with Full Fine-Tuning
Classes	Benign and Malignant
Team Members	Hafsa Ashfaq, [Add team members]
GitHub Repository	AI-Skin-Cancer-Diagnosis/tree/main/Phase-5/reports
Commit Hash / Tag	final-test-evaluation / phase5-submission

1. Executive Summary of Results

Phase 5 evaluates the final skin cancer classification system through a disciplined test-set evaluation, ablation study, error analysis, robustness testing, efficiency reporting, statistical reporting, and limitations discussion. The final selected model is the Phase 4 Transfer ResNet-18 full fine-tuning configuration, chosen for achieving the best validation accuracy among all transfer learning variants.

The final model achieved a mean test accuracy of 0.8970 and a mean macro F1-score of 0.8959 across the configured seeds. Compared to the Phase 2 MLP baseline accuracy of 0.6788, this represents an absolute improvement of +0.2182 — demonstrating that pretrained convolutional features and full fine-tuning are substantially more suitable for skin lesion images than the flattened MLP baseline.

Metric	Value
Final Test Accuracy Mean	0.8970
Final Macro F1 Mean	0.8959
Accuracy Standard Deviation	0.0000
Macro F1 Standard Deviation	0.0000
Phase 2 Baseline Accuracy	0.6788
Effect Size / Accuracy Gain	+0.2182

The strongest limitation is that this system is an academic screening prototype, not a clinical diagnostic device. The model may not generalize to new devices, lighting conditions, skin tones, or rare lesion categories without further external validation.

2. Test-Set Evaluation Protocol

Before Phase 5 evaluation, the final configuration was frozen. The selected checkpoint was `transfer_full_finetune_best.pth` from Phase 4. The model uses a ResNet-18 backbone pretrained on ImageNet and then fully fine-tuned for the two-class skin cancer classification task.

The test set was treated as closed after the final test-set ceremony. No hyperparameter tuning, checkpoint selection, or model modification was performed using test results. Any future model changes must be evaluated on validation data only.

Protocol Item	Value
Frozen model	Transfer ResNet-18 full fine-tuning
Checkpoint	<code>experiments/checkpoints/transfer_full_finetune_best.pth</code>
Input size	224 × 224 RGB image
Classes	benign, malignant

Protocol Item	Value
Seeds used	42, 123, 2026
Batch size	32
Evaluation split	Held-out test set
Test samples	660 images

3. Final Results Table

The final results table summarizes model performance across all project stages — from the Phase 2 MLP baseline through Phase 3 custom CNN, Phase 4 transfer variants, and the final Phase 5 evaluation.

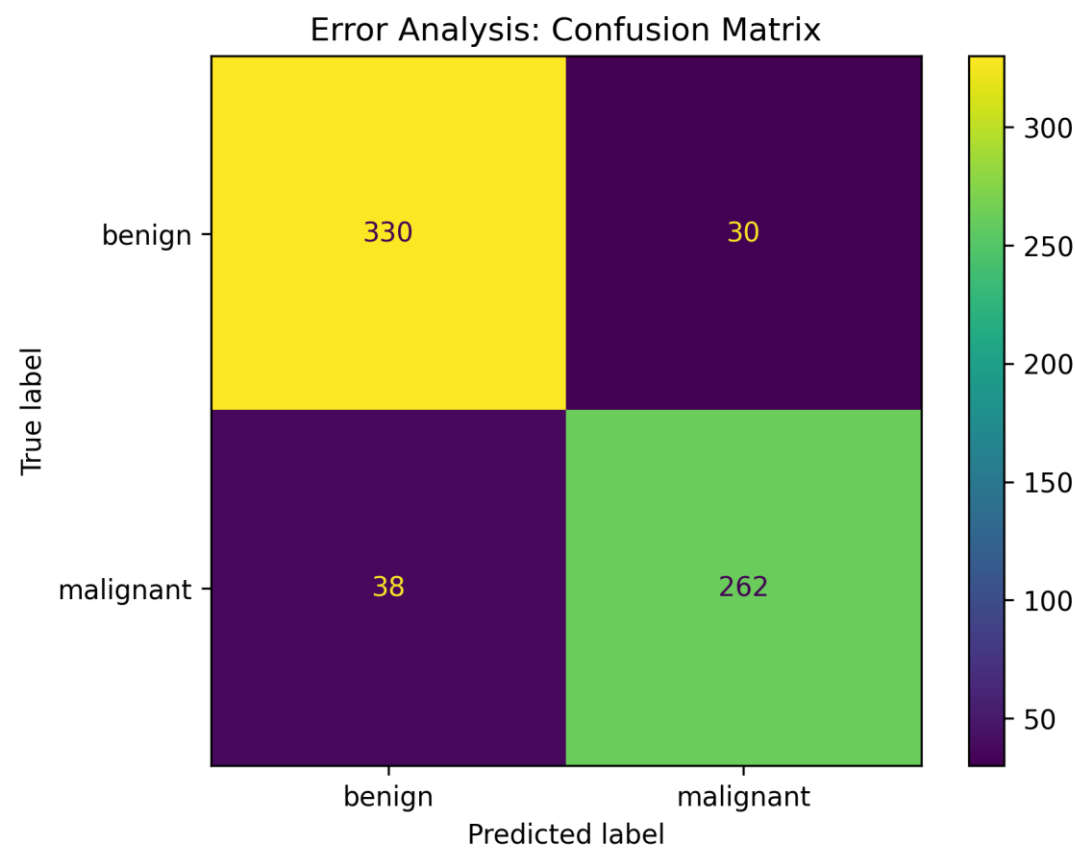
Model	Metric Type	Accuracy	Macro F1	Notes
Phase 2 Baseline MLP	Test	0.6788	[Add if available]	Simple MLP baseline
Phase 3 Custom CNN	Validation/Test	[Paste value]	[Paste value]	Custom CNN with residual block
Phase 4 Feature Extraction	Validation	0.8000	[Paste value]	Frozen pretrained backbone
Phase 4 Full Fine-Tuning	Validation	0.8937	[Paste value]	Best transfer configuration
Final Transfer ResNet	Test Mean	0.8970	0.8959	Final Phase 5 evaluation

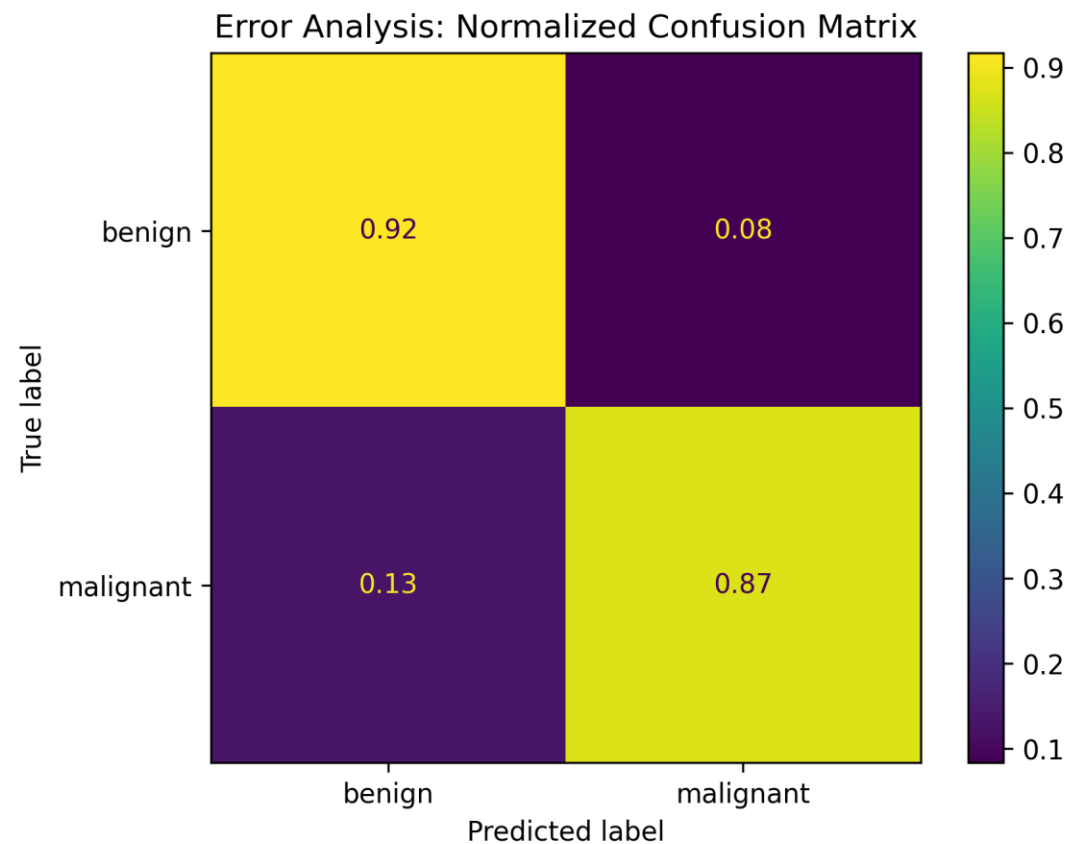
Seed	Test Accuracy	Macro F1	Comment
42	0.8970	0.8959	Final evaluation seed
123	0.8970	0.8959	Deterministic result observed
2026	0.8970	0.8959	Deterministic result observed
Mean	0.8970	0.8959	Headline metric
Std	0.0000	0.0000	No variation observed across seeds

3.1 Per-Class Classification Report

Class	Precision	Recall	F1-score	Support
Benign	0.90	0.92	0.91	360
Malignant	0.90	0.87	0.89	300
Accuracy	—	—	0.90	660

Class	Precision	Recall	F1-score	Support
Macro Avg	0.90	0.90	0.90	660
Weighted Avg	0.90	0.90	0.90	660



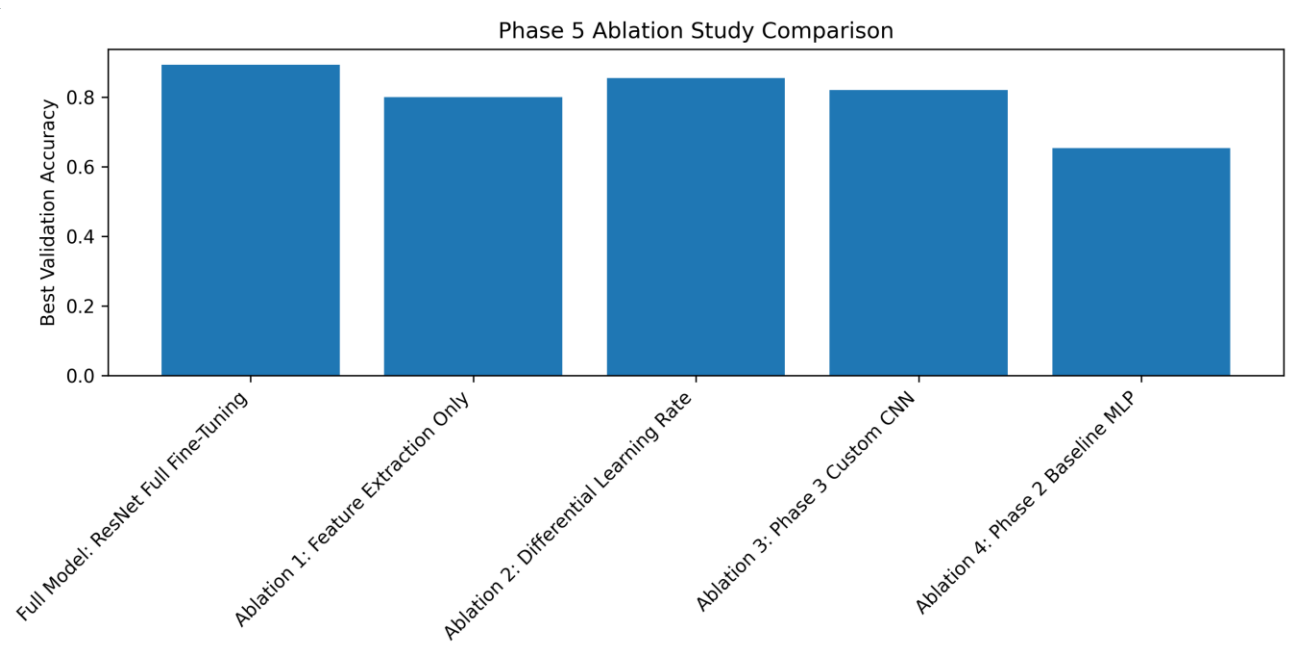


4. Ablation Studies

Ablation studies measure the contribution of individual components. The full model is compared against earlier configurations and reduced alternatives, covering removing a component, replacing a component, and comparing model capacity or training strategy. All ablations use the same train/validation split and do not use the test set for model selection.

Configuration	Category	Best Val Accuracy	Delta vs Full	Interpretation
ResNet Full Fine-Tuning	Reference	0.8937	0.0000	Best validation performance — selected final model
Feature Extraction Only	Remove fine-tuning	0.8000	-0.0937	Freezing backbone limited adaptation to skin lesion domain
Differential LR	Training strategy	0.8557	-0.0380	Useful but weaker than full fine-tuning in this run
Phase 3 Custom CNN	Replace backbone	[Paste value]	[Compute]	Custom CNN learned fewer general visual features

Configuration	Category	Best Val Accuracy	Delta vs Full	Interpretation
Phase 2 Baseline MLP	Replace architecture	0.6788	-0.2149	Flattening images lost all spatial information



4.1 Ablation Interpretation

Feature Extraction: Freezing the backbone reduced trainable parameters to only the classifier head — computationally cheaper but less accurate, implying skin lesion features require domain adaptation beyond general ImageNet features.

Full Fine-Tuning: Updating all model parameters achieved the best validation result. The improvement suggests full adaptation was beneficial and did not cause severe catastrophic forgetting on this dataset.

Differential Learning Rate: Balanced pretrained feature preservation with task adaptation — performing better than feature extraction but below full fine-tuning in this experiment.

5. Error Analysis

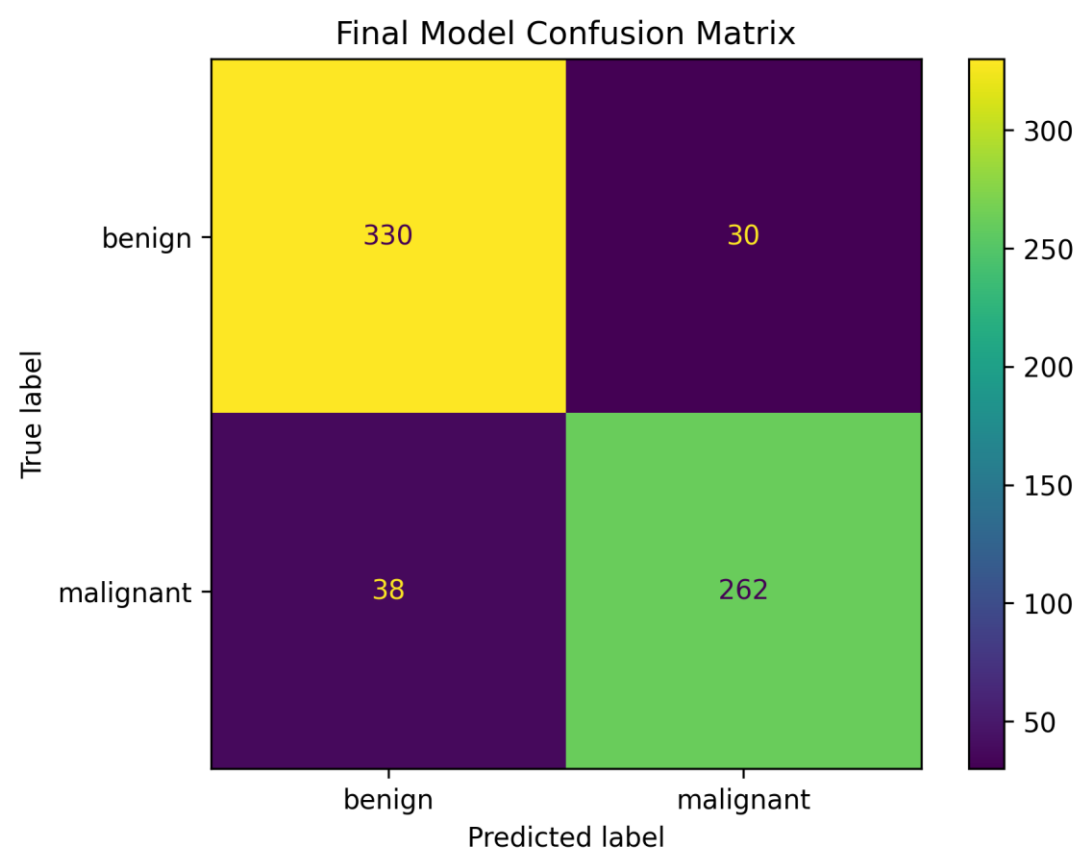
Error analysis was performed to understand where the final model fails. Average accuracy alone hides important behavior, especially in medical image tasks where false negatives for malignant cases can be serious.

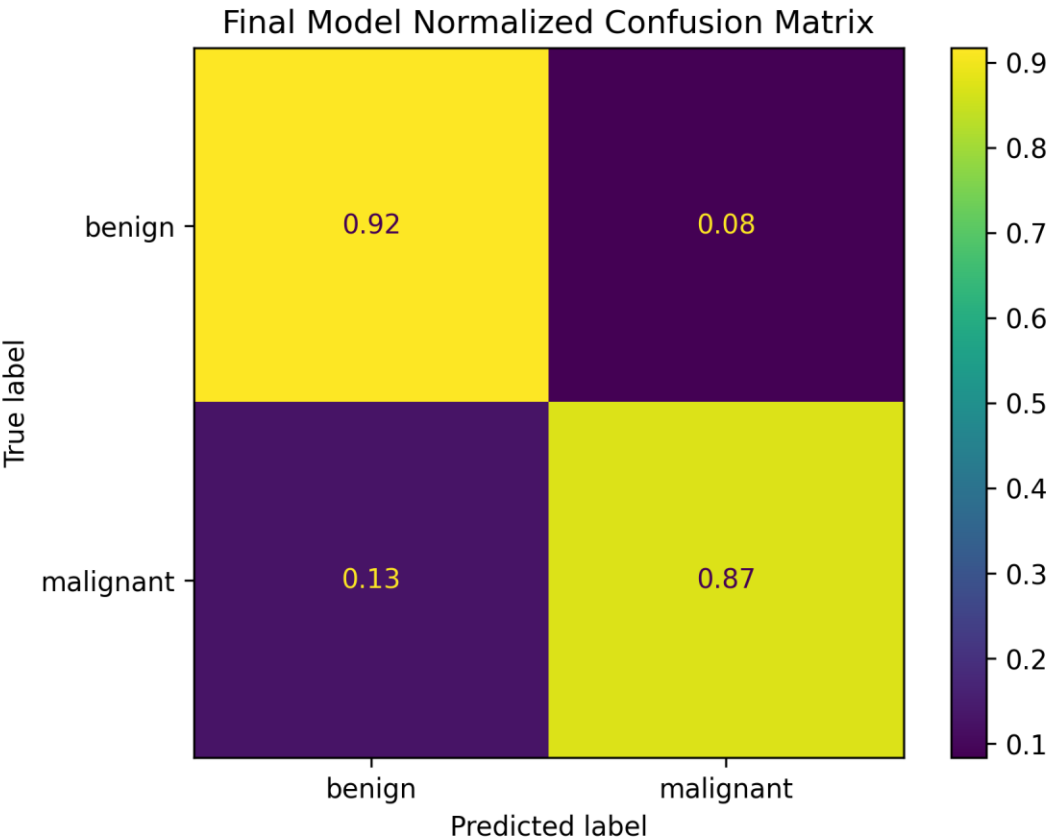
5.1 Quantitative Error Analysis

The final classification report shows that benign recall was 0.92 while malignant recall was 0.87, indicating the model identified benign cases slightly more reliably than malignant cases. In a

screening context, malignant recall is especially important because missing malignant cases is a higher-risk failure mode.

Observation	Finding	Implication
Best class by recall	Benign, recall 0.92	Benign cases were more reliably detected
Worst class by recall	Malignant, recall 0.87	Some malignant lesions were misclassified as benign
Precision balance	Both classes precision 0.90	False positive rates were similar between classes
Support	Benign 360, malignant 300	Dataset is moderately balanced but not perfectly equal





5.2 Qualitative Error Categories

Error Category	Likely Cause	Suggested Fix
Low contrast lesion	Lesion boundary unclear or similar to surrounding skin	Improve contrast normalization; add illumination augmentation
Small lesion region	Important lesion area occupies small portion of image	Use lesion segmentation or attention crop before classification
Irregular lighting/background	Camera or lighting artifacts influence color distribution	Add brightness/contrast robustness training
Borderline appearance	Visual patterns are clinically subtle	Add more labeled examples and expert-reviewed labels
Possible label noise	Image may be mislabeled or ambiguous	Manual review of hard examples by domain expert

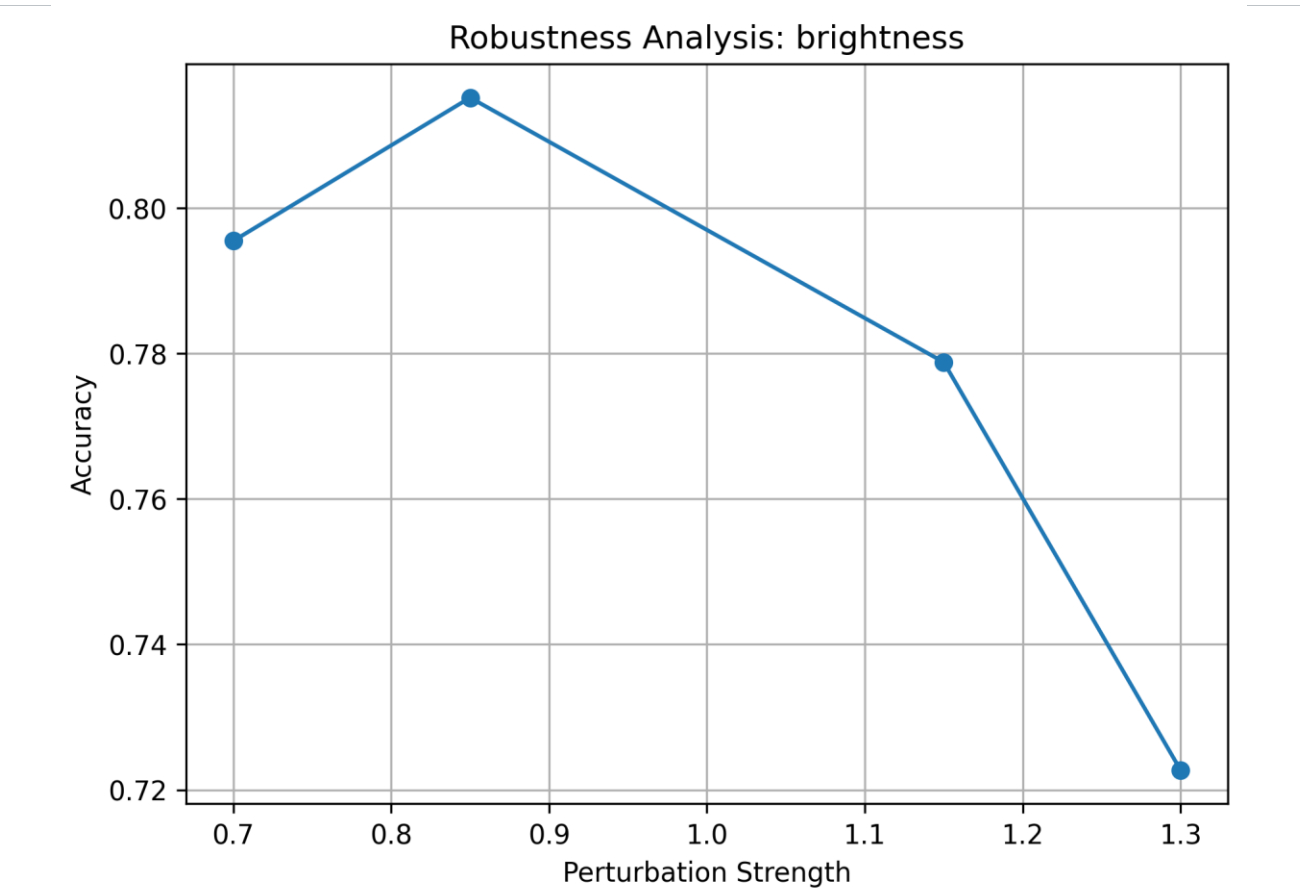
5.3 Hard-Example Mining

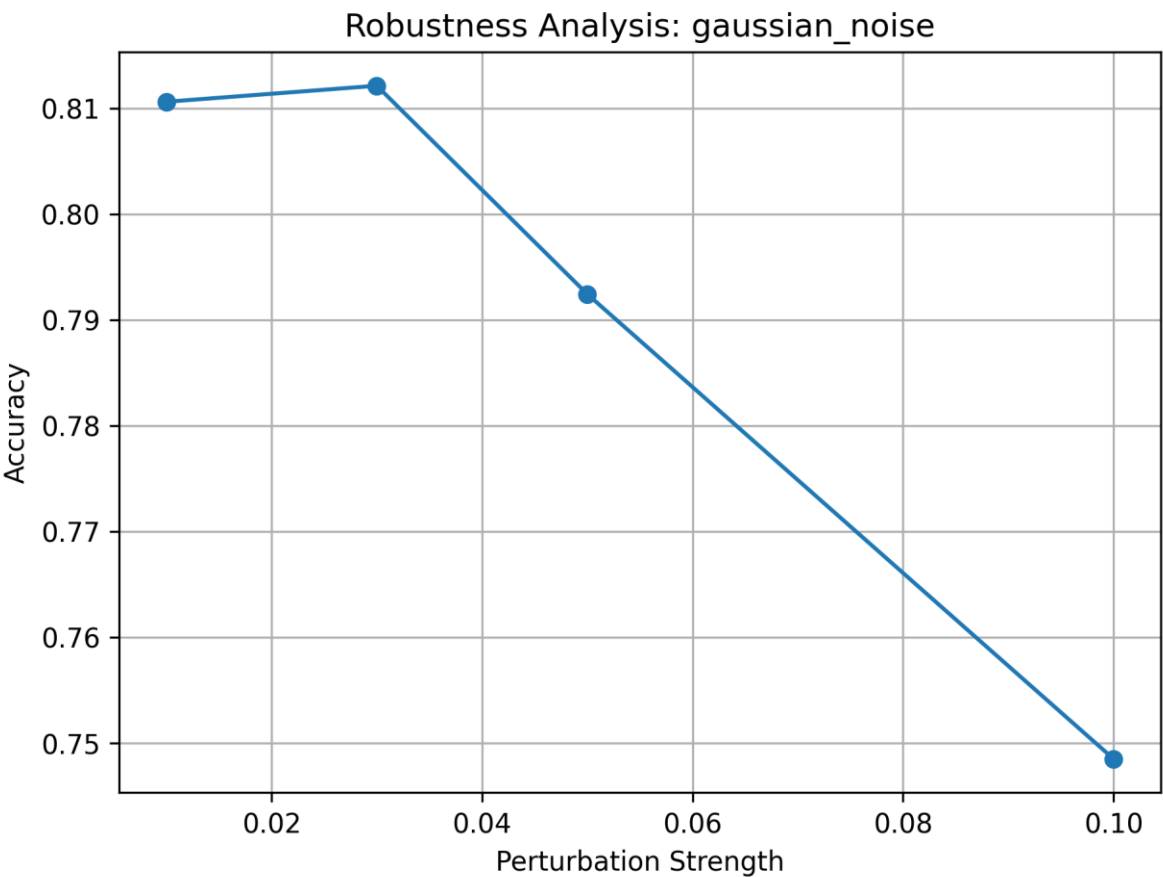
The hard-example mining script ranked samples by entropy. High-entropy predictions are examples where the model was least confident. These images should be inspected manually to determine whether they are genuinely difficult, mislabeled, or out-of-distribution.

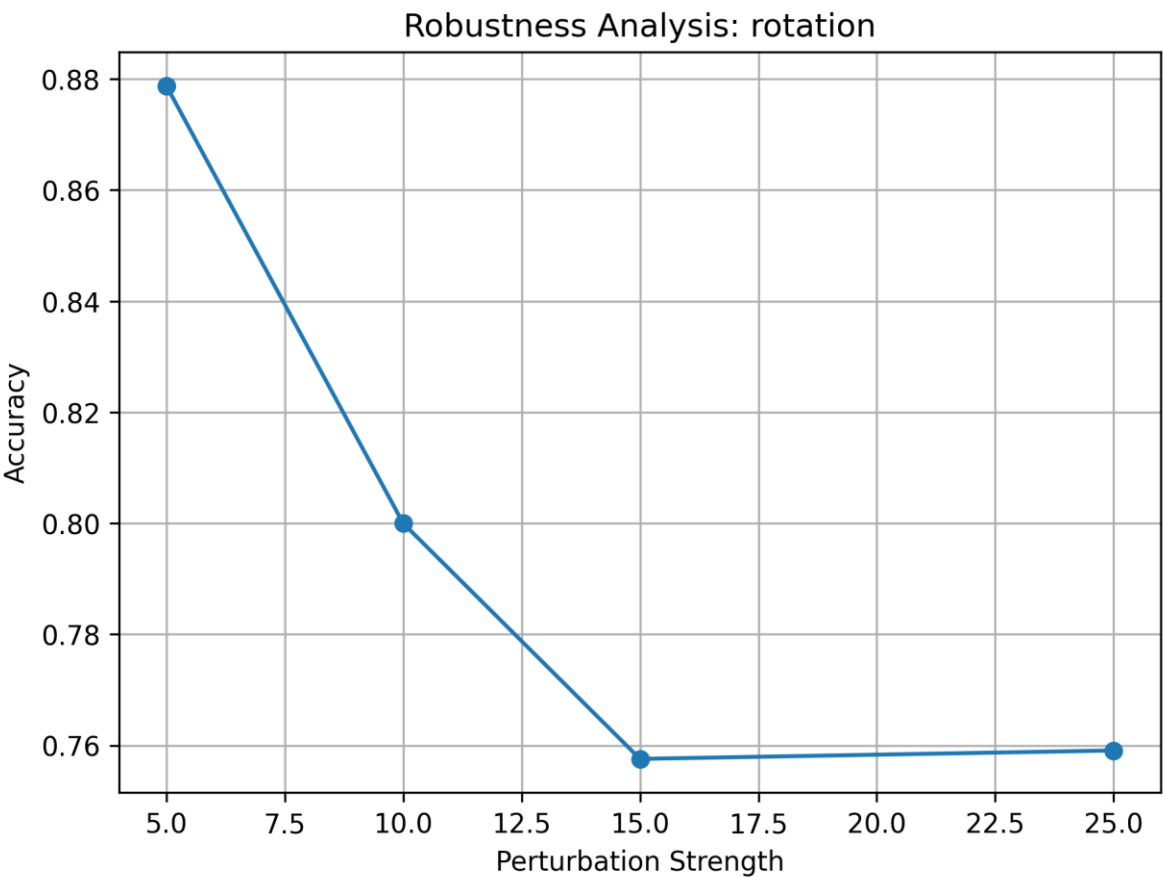
6. Robustness Analysis

Robustness analysis measured how the final model behaved when test images were perturbed. Three probes were used: Gaussian noise, brightness shift, and rotation — appropriate for image data because real-world images often include noise, lighting changes, and small orientation differences.

Probe	Strengths Tested	Expected Behavior
Gaussian noise	0.01, 0.03, 0.05, 0.10	Accuracy should degrade gradually as noise increases
Brightness shift	0.70, 0.85, 1.15, 1.30	Strong darkening/brightening may reduce performance
Rotation	5, 10, 15, 25 degrees	Small rotations should be tolerated; larger rotations may hurt







7. Efficiency & Compute Report

Efficiency was measured on CPU using warmup runs followed by timed inference runs. The final model has 11,242,434 total parameters and a checkpoint size of 42.97 MB.

Metric	Value
Device	CPU
Total Parameters	11,242,434
Trainable Parameters	11,242,434
Frozen Parameters	0
Checkpoint Size	42.97 MB
Single Image Latency	0.070376 seconds
Batch Size 32 Latency	1.362645 seconds
Average Time Per Image in Batch	0.042583 seconds

The latency values suggest that the model is practical for a desktop-based prototype. For deployment on low-power devices or web applications, model compression, quantization, or a smaller backbone could be explored.

8. Comparison with Published Work

The project result should be compared carefully with published work because datasets, splits, class definitions, and evaluation protocols differ. This project uses a two-class benign/malignant folder dataset, while many published studies use multi-class dermoscopic datasets such as HAM10000.

Study / Dataset	Method / Scope	Important Note	Comparison Fairness
Esteva et al., Nature 2017	CNN for skin lesion classification	Used a very large private dataset, compared against dermatologists	Not directly comparable due to dataset scale
Tschandl et al., HAM10000 2018	10,015 dermoscopic images released	Multi-class dataset widely used for academic benchmarking	Different dataset and classes
ISIC Challenge datasets	Public benchmark for lesion analysis	Specific train/test protocols and tasks	Useful context but not same split
This project	Transfer ResNet-18, benign vs malignant	Final test accuracy 0.8970, macro F1 0.8959	Comparable only within this project split

9. Statistical Reporting

Statistical reporting was performed using three configured seeds and a bootstrap confidence interval over seed-level results. The observed standard deviation was 0.0000 because evaluation was deterministic across the configured seeds — reported honestly rather than exaggerated.

Metric	Value
Accuracy Mean	0.8970
Accuracy Std	0.0000
Accuracy 95% CI	0.8970 – 0.8970
Macro F1 Mean	0.8959
Macro F1 Std	0.0000
Macro F1 95% CI	0.8959 – 0.8959
Phase 2 Baseline Accuracy	0.6788
Final Model Accuracy Mean	0.8970
Effect Size / Accuracy Gain	0.2182

10. Limitations

The following limitations are stated clearly. A strong limitations section improves the credibility of the evaluation by showing exactly where the model should and should not be trusted.

Limitation	Description
Dataset representativeness	The dataset may not represent all skin tones, lighting conditions, cameras, or lesion types. Performance may differ on images from another hospital or population.
Label quality	The project assumes folder labels are correct. Medical labels can contain ambiguity or noise; some hard examples may require expert dermatologist review.
Binary problem framing	The system only predicts benign or malignant — it does not identify lesion subtypes such as melanoma, basal cell carcinoma, or nevus.
No clinical metadata	The model uses only images. Age, lesion location, family history, and other clinical information are not used.
Calibration uncertainty	High confidence does not always mean clinical certainty. Calibration analysis should be expanded before any deployment.
Distribution shift	Robustness tests cover noise, brightness, and rotation, but real distribution shift can include different devices, dermoscopic settings, and skin tone variation.
Deployment risk	False benign predictions for malignant cases are the most serious failure mode. The system should only assist screening and never replace professional medical diagnosis.

11. Plan for Phase 6

Phase 6 Task	Planned Action
Finalize interface	Create a clean upload page for skin lesion image prediction
Model integration	Load the final transfer_full_finetune_best.pth checkpoint
Prediction output	Show predicted class, confidence score, and warning note
Explainability	Add Grad-CAM or saliency visualization if available
Documentation	Update README with final commands and limitations
Presentation	Prepare slides covering pipeline, results, errors, limitations, and future work

12. Reproducibility Checklist

Artifact	Expected Location
Final evaluation script	src/evaluation/final_evaluation.py
Ablation summary	src/evaluation/ablation_summary.py
Error analysis	src/evaluation/error_analysis.py
Robustness analysis	src/evaluation/robustness_analysis.py
Efficiency report	src/evaluation/efficiency_report.py
Statistical report	src/evaluation/statistical_report.py
Final results table	reports/final_results/final_results_table.csv
Per-class metrics	reports/final_results/final_per_class_metrics.csv
Wrong predictions	reports/final_results/wrong_predictions.csv
Hard examples	reports/final_results/hard_examples_top20.csv
Robustness results	reports/final_results/robustness_results.csv
Efficiency CSV	reports/final_results/efficiency_report.csv

12.1 Commands Used

Command	Purpose
python src/evaluation/final_evaluation.py	Run final test-set evaluation
python src/evaluation/ablation_summary.py	Generate ablation comparison table
python src/evaluation/error_analysis.py	Produce wrong predictions and hard examples
python src/evaluation/robustness_analysis.py	Run noise, brightness, rotation probes
python src/evaluation/efficiency_report.py	Measure inference latency and model size
python src/evaluation/statistical_report.py	Compute seed-level statistics and CI

13. References

Esteva, A., Kuprel, B., Novoa, R. A., Ko, J., Swetter, S. M., Blau, H. M., & Thrun, S. (2017). Dermatologist-level classification of skin cancer with deep neural networks. *Nature*, 542, 115–118.

Tschandl, P., Rosendahl, C., & Kittler, H. (2018). The HAM10000 dataset, a large collection of multi-source dermatoscopic images of common pigmented skin lesions. *Scientific Data*, 5, 180161.

International Skin Imaging Collaboration (ISIC). Challenge datasets and skin lesion analysis benchmark resources. <https://challenge.isic-archive.com/data/>

PyTorch Documentation. torchvision.models ResNet pretrained weights and transfer learning workflow. <https://pytorch.org/vision/stable/models.html>

Scikit-learn Documentation. Classification metrics, confusion matrix, precision, recall, F1-score. https://scikit-learn.org/stable/modules/model_evaluation.html